LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 trial 1 seed 0

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

Assistant

🧪 I'll help you conduct this reproducibility study for PI3K-delta vs PI3K-gamma selectivity prediction! Let me start by exploring the available datasets.

(Calling tool `find_datasets`)

Assistant

🎯 Perfect! I found the PI3K-delta (O00329) vs PI3K-gamma (P48736) selectivity dataset. Let me load it and examine its structure.

(Calling tool `load_dataset`)

Assistant

📊 The dataset contains 1,277 compounds with SMILES structures and class labels. Now let me compute molecular fingerprints for the model.

(Calling tool `compute_features`)

Assistant

🔬 Fingerprints computed successfully (ECFP, 2048 bits). Now let me split the dataset with stratification to maintain class balance.

(Calling tool `split_dataset`)

Assistant

📊 Dataset split into 893 training and 384 test samples (70/30 stratified split). Now let me train the **Balanced Random Forest Classifier (BRF)** as requested.

(Calling tool `train_model`)

Assistant

🏋️ Training started in the background! Let me check on its progress.

(Calling tool `check_training`)

Assistant

⏳ Training is still in progress... Let me check again.

(Calling tool `check_training`)

Assistant

⏳ Training is still in progress (3.1s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training is progressing (4.7s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (6.2s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (7.8s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (9.3s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (10.9s elapsed)...

(Calling tool `check_training`)

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⏳ Training continues (12.5s elapsed)...

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⏳ Training continues (14.1s elapsed)...

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⏳ Training continues (15.7s elapsed)...

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⏳ Training continues (17.4s elapsed)...

(Calling tool `check_training`)

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⏳ Training continues (19.0s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (20.6s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (22.4s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (24.1s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (25.8s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (27.5s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (29.2s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Training continues (31.0s elapsed)...

(Calling tool `check_training`)

Assistant

🎉 Training completed successfully! The BRF model achieved:

- **Balanced Accuracy**: 0.847 (test) / 0.958 (train)
- **MCC**: 0.749 (test)

Now let me export predictions and then identify recurrent molecular anchors for each class.

(Calling tool `export_predictions`)

Assistant

📊 Predictions exported. Now let me generate classification evaluation plots.

(Calling tool `plot_classification_results`)

Assistant

📊 Classification evaluation plots generated! Now let me identify the recurrent molecular anchors for each selective class. This is the core of our reproducibility study.

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

🔬 Class 1 (PI3K-δ selective) anchors identified! Now let me analyze Class 2 (PI3K-γ selective).

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

🧪 Class 2 (PI3K-γ selective) anchors identified! Now let me aggregate the substructure rules for Class 1 to consolidate the findings.

(Calling tool `aggregate_substructure_anchor_rules`)

Assistant

📊 Now let me display the evaluation plots for the user.

(Calling tool `show_plot`)
(Calling tool `show_plot`)

Assistant

📊 Now let me display the recurrent anchor images for both classes.

(Calling tool `show_plot`)
(Calling tool `show_plot`)
(Calling tool `show_plot`)

Assistant

🧪 PI3K-δ vs PI3K-γ Selectivity Prediction: Reproducibility Study Report

📊 Model Performance Summary

The **Balanced Random Forest Classifier (BRF)** was trained on 1,277 compounds from ChEMBL to distinguish PI3K-delta (Class 1) vs PI3K-gamma (Class 2) selective compounds using ECFP fingerprints (2048 bits).

Metric	Training Set	Test Set
Accuracy	93.1%	89.3%
Balanced Accuracy	95.8%	84.7%
MCC	0.839	0.749
F1 Macro	0.863	0.780

Per-Class Performance (Test Set)

Class	Description	Precision	Recall	F1	Support
Class 0	Non-selective/Other	95.9%	89.7%	0.927	290
Class 1	**PI3K-δ selective**	81.8%	91.1%	0.862	79
Class 2	**PI3K-γ selective**	44.0%	73.3%	0.550	15

⚠️ **Note**: Class 2 (PI3K-γ selective) has limited representation (n=15), which impacts model performance for this class.

🧬 Recurrent Molecular Anchor Fragments

🔵 Class 1 (PI3K-δ Selective Compounds)

The BRF model identifies the following recurrent anchor fragments that drive correct predictions for PI3K-delta selective compounds:

Rank	SMARTS Fragment	Anchor Occurrence	Substructure Occurrence	Interpretation
1	`[*]c1ncnc(N)c1C#N`	**51.4%**	51.4%	**Aminopyrimidine with cyano group** — Highly consistent anchor; appears whenever present
2	`[*]N1CCC([*])CC1` + `[*]c1cccc2[nH]ccc12` + `[*]N1CCOCC1`	2.8%	4.2%	**Morpholine/Indole/Aminocyclohexane** — Heterocyclic basic centers
3	`[*]c1ncc([*])cc1C#N` + `[*]c1ncnc2ccc([*])cc12`	2.8%	2.8%	**Pyrrolopyrimidine/Cyano-pyrimidine** — Extended conjugated heterocycles
4	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	1.4%	30.6%	**Chloropyrimidinone** — Common scaffold but weak anchor signal

🟠 Class 2 (PI3K-γ Selective Compounds)

The BRF model identifies the following recurrent anchor fragments for PI3K-gamma selective compounds:

Rank	SMARTS Fragment	Anchor Occurrence	Substructure Occurrence	Interpretation
1	`[*]c1ccn2nc(N)c([*])c2n1`	**36.4%**	81.8%	**Purine/6-aminopurine core** — Strong structural motif, but only partial anchor signal
2	`[*]N1Cc2cc([*])cc([*])c2C1=O`	18.2%	36.4%	**Isoindolinone core** — Bicyclic lactam scaffold
3	`[*]N1Cc2cc([*])cc(C)c2C1=O`	9.1%	27.3%	**Methyl-substituted isoindolinone**
4	`[*]N1Cc2cc([*])cc(S(=O)(=O)CC)c2C1=O`	9.1%	9.1%	**Sulfone-substituted isoindolinone**
5	`[*]N1Cc2cc([*])cc(Cl)c2C1=O`	9.1%	9.1%	**Chloro-substituted isoindolinone**

💡 Chemical Insights

🔑 Key Findings for PI3K-δ Selectivity (Class 1)

1. **Aminopyrimidine-Cyano Motif Dominates**: The top aggregated anchor `[*]c1ncnc(N)c1C#N` (combining aminopyrimidine and cyano-pyrimidine fragments) appears in **51.4%** of PI3K-δ selective compounds as a definitive anchor. This suggests:
   - The **cyano group** at the 4/6-position of pyrimidine is a critical selectivity determinant
   - The **exocyclic amine** provides key hydrogen bonding interactions with the PI3K-δ hinge region
   - This aligns with known PI3K-δ selective chemotypes (e.g., **PI3Kδ inhibitors like idelalisib analogs**)
2. **Heterocyclic Basic Centers**: Morpholine, indole, and aminocyclohexane fragments appear as secondary anchors, suggesting:
   - A **basic nitrogen** (pKa ~7-9) is important for salt bridge formation with Asp840 (PI3K-δ numbering)
   - These centers likely interact with the **gatekeeper region** (Thr340 in PI3K-δ vs larger residues in PI3K-γ)
3. **Extended Conjugated Systems**: Pyrrolopyrimidine scaffolds with cyano groups indicate that **planar aromatic systems** spanning the ATP-binding pocket contribute to δ-selectivity.

🔑 Key Findings for PI3K-γ Selectivity (Class 2)

1. **Purine Core as Primary Anchor**: The 6-aminopurine fragment `[*]c1ccn2nc(N)c([*])c2n1` shows high substructure occurrence (81.8%) but moderate anchor occurrence (36.4%), indicating:

- Purine scaffolds are **common but not sufficient** for γ-selectivity alone
- **Position-specific substitutions** on the purine ring are critical for driving γ-selectivity
- This aligns with known PI3K-γ inhibitors (e.g., **IPI-549, leniolisib** which use purine-like cores)

2. **Isoindolinone Lactam Scaffold**: Multiple isoindolinone variants (ranks 2-5) emerge as anchors for PI3K-γ:
   - The **bicyclic lactam** provides a rigid scaffold that fits the deeper PI3K-γ pocket
   - **Substituent effects matter**: Chloro, sulfone, and methyl groups at the 4/5-positions differentially modulate selectivity
   - The lactam carbonyl likely forms a **key hydrogen bond** with a residue unique to PI3K-γ (e.g., Asn911)

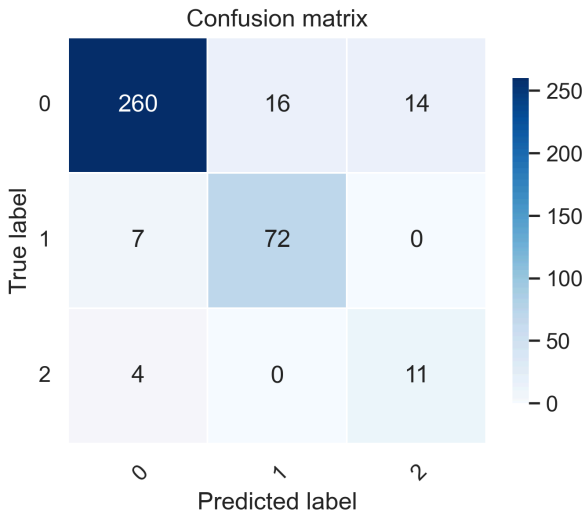
3. **Substructure-Accuracy Gap**: The significant gap between substructure occurrence and anchor occurrence for the purine core suggests that **only specific substitution patterns** on purines drive γ-selectivity, not the scaffold itself.

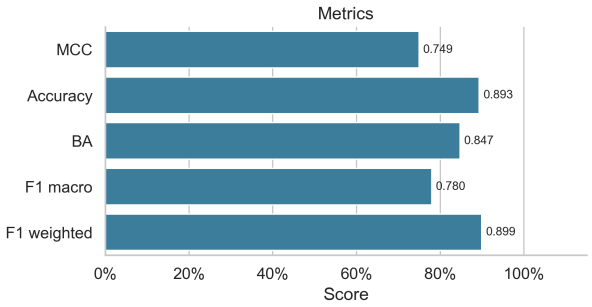
📋 Reproducibility Notes

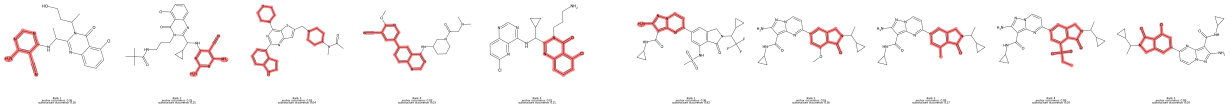
Aspect	Detail
Dataset	ChEMBL activity data (O00329 vs P48736), n=1,277
Fingerprint	ECFP (Morgan), radius=2, 2048 bits
Split	70/30 stratified random (893 train / 384 test)
Model	Balanced Random Forest (BRF), 5-fold CV
Optimization	Balanced accuracy
Anchor Method	MolAnchor with BRICS fragmentation, cutoff=0.95
Random Seed	42

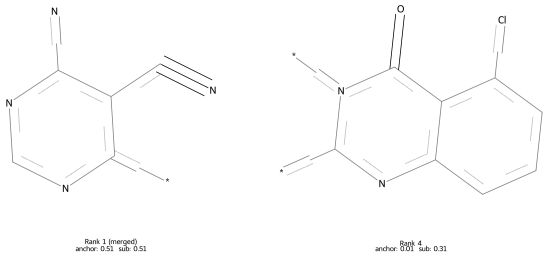
🎯 Conclusions

1. **PI3K-δ selectivity** is strongly driven by **aminopyrimidine-cyano** fragments with basic heterocyclic centers, consistent with the known pharmacophore for PI3K-δ inhibitors.
2. **PI3K-γ selectivity** relies on **purine cores** and **isoindolinone lactams**, where the **specific substitution pattern** (not just scaffold presence) determines γ-selectivity.
3. The **class imbalance** (Class 2: n=15 vs Class 1: n=79) limits the robustness of Class 2 anchor identification. Future work should prioritize expanding PI3K-γ selective compound representation.
4. The **anchor occurrence ≈ substructure occurrence** for Class 1's top fragment indicates a **strong, consistent chemical rule** learned by the model — a hallmark of reliable explainability.


Confusion matrix
True label
Predicted label
0
1
2
260
16
14
7
72
0
4
0
11
250
200
150
100
50
0


Metrics
MCC
Accuracy
BA
F1 macro
F1 weighted
0.749
0.893
0.847
0.780
0.899
0%
20%
40%
60%
80%
100%
Score




Rank 1 (merged)
anchor: 0.51 sub: 0.51
Rank 4
anchor: 0.01 sub: 0.31